DSA Assignment 1

Code:

1) File - array.h

```
typedef struct array{
    int* A;
    int size;
    int len;
}array;

void init(array* a, int size);
int is_empty(array* a);
int is_full(array* a);
void append(array* a, int data);
void insert_at_index(array* a, int idx, int data);
void remove_at_index(array* a, int idx);
void display(array* a);
int max(array* a);
int min(array* a);
void reverse(array* a);
void merge(array* a1, array* a2, array* res);
```

2) File - array.c

```
#include <stdio.h>
#include <stdlib.h>
#include <limits.h>

#include "array.h"

void init(array* a, int size){
    a -> size = size;
    a -> len = 0;
    a -> A = (int*) malloc(sizeof(int) * size);
    return;
}
```

```
int is_empty(array* a){
    return a -> len == 0;
}

int is_full(array* a){
    return a -> size == a -> len;
}

void append(array* a,int data){
    if(is_full(a)){
        return;
    }
    a -> A[a -> len++] = data;
    return;
}

void insert_at_index(array* a, int idx, int data){
    if(is_full(a)){
        return;
    }
    for(int i = a->len; i >= idx; i--){
        a -> A[i + 1] = a -> A[i];
    }
    a -> A[idx] = data;
    a -> len++;
    return;
}

void remove_at_index(array* a, int idx){
    if (is_empty(a)){
        return;
    }
    for(int i=idx; i < a->len; i++){
        a->A[i] = a->A[i+1];
    }
    a -> len--;
    return;
}
```

```
void display(array* a){
    if (is_empty(a)){
            return;
    }
    for (int i = 0; i < a->len; i++){
            printf("%d ", a->A[i]);
    }
    printf("\n");
    return;
}

int max(array* a){
    if (is_empty(a)){
            return INT_MIN;
    }
    int ans = INT_MIN;
    for (int i=0; i < a->len; i++){
            if(ans < a -> A[i]){
                    ans = a -> A[i];
            }
    }
    return ans;
}

int min(array* a){
        if (is_empty(a)){
                return INT_MIN;
        }
        int ans = INT_MAX;
        for (int i = 0; i < a->len; i++){
                if (ans > a->A[i]){
                ans = a -> A[i];
                }
        }
        return ans;
}


void reverse(array* a){
    if (is_empty(a)){
```

```
        return;
    }
    int start = 0;
    int end = a->len - 1;
    while(start < end){
        int temp = a->A[start];
        a->A[start] = a->A[end];
        a->A[end] = temp;
        start++;
        end--;
    }
    return;
}

void merge(array* a1, array* a2, array* res){

    for (int i = 0; i < a1->len; i++){
        res->A[i] = a1->A[i];
        res->len++;
    }
    for (int j = 0; j < a2->len; j++){
        res->A[res->len++] = a2->A[j];
    }
}
```

3) File - main.c

```
#include "array.c"
#include <limits.h>

int main()
{
	array a, a1, a2, a3;
	int choice, val, ind, c = 1, size, len;

	while(c) {
	printf("Enter 1 to initialize the array\n");
	printf("Enter 2 to append an element to the array\n");
	printf("Enter 3 to insert an element at a particular index\n");
	printf("Enter 4 to remove the element at a particular index\n");
	printf("Enter 5 to display the elements of the array\n");
	printf("Enter 6 to display the maximum element of the array\n");
	printf("Enter 7 to display the minimum element of the array\n");
	printf("Enter 8 to merge two arrays\n");

	scanf("%d", &choice);

	switch(choice) {
	case 1:
		printf("Enter the maximum size of the array: ");
		scanf("%d", &size);
		printf("Enter the number of elements to populate the array with: ");
		scanf("%d", &len);

		init(&a, size);
		for(int i = 0; i < len; i++)
		append(&a, rand());

		break;
	case 2:
		printf("Enter the element to be appended to the array: ");
		scanf("%d", &val);
		append(&a, val);
		break;
	case 3:
```

```
        printf("Enter the value to be inserted into the array: ");
        scanf("%d", &val);
        printf("Enter the index at which the value is to be inserted: ");
        scanf("%d", &ind);
        insert_at_index(&a, ind, val);
        break;
case 4:
        printf("Enter the index of the element which is to be removed: ");
        scanf("%d", &ind);
        remove_at_index(&a, ind);
        break;
case 5:
        display(&a);
        break;
case 6:
        printf("Maximum element of the array: %d\n", max(&a));
        break;
case 7:
        printf("Minimum element of the array: %d\n", min(&a));
        break;
case 8:
        init(&a1, 2);
        append(&a1, 1);
        append(&a1, 2);
        printf("First array: ");
        display(&a1);

        init(&a2, 2);
        append(&a2, 3);
        append(&a2, 4);
        printf("Second array: ");
        display(&a2);

        init(&a3, 4);
        merge(&a1, &a2, &a3);
        printf("Merged array: ");
        display(&a3);

        break;
default:
```

```
                break;
        }

        printf("Enter 1 to continue or 0 to stop: ");
        scanf("%d", &c);
        printf("\n");
        }
        return 0;
}
```

Output:

```
Terminal                                                    Aug 22  00:05
                                                arnav@arnav-desktop-1: ~/Deskt
arnav@arnav-desktop-1:~/Desktop/Assignment 1$ gcc main.c -o main -Wall
arnav@arnav-desktop-1:~/Desktop/Assignment 1$ ./main
Enter 1 to initialize the array
Enter 2 to append an element to the array
Enter 3 to insert an element at a particular index
Enter 4 to remove the element at a particular index
Enter 5 to display the elements of the array
Enter 6 to display the maximum element of the array
Enter 7 to display the minimum element of the array
Enter 8 to merge two arrays
1
Enter the maximum size of the array: 5
Enter the number of elements to populate the array with: 4
Enter 1 to continue or 0 to stop: 1

Enter 1 to initialize the array
Enter 2 to append an element to the array
Enter 3 to insert an element at a particular index
Enter 4 to remove the element at a particular index
Enter 5 to display the elements of the array
Enter 6 to display the maximum element of the array
Enter 7 to display the minimum element of the array
Enter 8 to merge two arrays
5
1804289383 846930886 1681692777 1714636915
Enter 1 to continue or 0 to stop: 1

Enter 1 to initialize the array
Enter 2 to append an element to the array
Enter 3 to insert an element at a particular index
Enter 4 to remove the element at a particular index
Enter 5 to display the elements of the array
Enter 6 to display the maximum element of the array
Enter 7 to display the minimum element of the array
Enter 8 to merge two arrays
2
Enter the element to be appended to the array: 1000
Enter 1 to continue or 0 to stop: 1
```

```
Enter 1 to initialize the array
Enter 2 to append an element to the array
Enter 3 to insert an element at a particular index
Enter 4 to remove the element at a particular index
Enter 5 to display the elements of the array
Enter 6 to display the maximum element of the array
Enter 7 to display the minimum element of the array
Enter 8 to merge two arrays
5
1804289383 846930886 1681692777 1714636915 1000
Enter 1 to continue or 0 to stop: 1

Enter 1 to initialize the array
Enter 2 to append an element to the array
Enter 3 to insert an element at a particular index
Enter 4 to remove the element at a particular index
Enter 5 to display the elements of the array
Enter 6 to display the maximum element of the array
Enter 7 to display the minimum element of the array
Enter 8 to merge two arrays
4
Enter the index of the element which is to be removed: 1
Enter 1 to continue or 0 to stop: 1

Enter 1 to initialize the array
Enter 2 to append an element to the array
Enter 3 to insert an element at a particular index
Enter 4 to remove the element at a particular index
Enter 5 to display the elements of the array
Enter 6 to display the maximum element of the array
Enter 7 to display the minimum element of the array
Enter 8 to merge two arrays
3
Enter the value to be inserted into the array: 5000
Enter the index at which the value is to be inserted: 3
Enter 1 to continue or 0 to stop: 1

Enter 1 to initialize the array
Enter 2 to append an element to the array
Enter 3 to insert an element at a particular index
Enter 4 to remove the element at a particular index
Enter 5 to display the elements of the array
```

```
Enter 1 to initialize the array
Enter 2 to append an element to the array
Enter 3 to insert an element at a particular index
Enter 4 to remove the element at a particular index
Enter 5 to display the elements of the array
Enter 6 to display the maximum element of the array
Enter 7 to display the minimum element of the array
Enter 8 to merge two arrays
5
1804289383 1681692777 1714636915 5000 1000
Enter 1 to continue or 0 to stop: 1

Enter 1 to initialize the array
Enter 2 to append an element to the array
Enter 3 to insert an element at a particular index
Enter 4 to remove the element at a particular index
Enter 5 to display the elements of the array
Enter 6 to display the maximum element of the array
Enter 7 to display the minimum element of the array
Enter 8 to merge two arrays
6
Maximum element of the array: 1804289383
Enter 1 to continue or 0 to stop: 1

Enter 1 to initialize the array
Enter 2 to append an element to the array
Enter 3 to insert an element at a particular index
Enter 4 to remove the element at a particular index
Enter 5 to display the elements of the array
Enter 6 to display the maximum element of the array
Enter 7 to display the minimum element of the array
Enter 8 to merge two arrays
8
First array: 1 2
Second array: 3 4
Merged array: 1 2 3 4
Enter 1 to continue or 0 to stop: 0

arnav@arnav-desktop-1:~/Desktop/Assignment 1$
```